

CSC 561 Individual Project Proposal

Project Title

Comparing Deep Learning Models for Hotel Demand Forecasting and Dynamic Room Pricing

Purpose, Motivation, Impact

The goal of this project is to build a machine learning system that can forecast hotel room demand and generate room pricing recommendations using real historical hotel data. Many hotels still rely on manual pricing decisions, spreadsheets, or simple fixed rules, which can be slow to update and may not respond well to changing demand. In contrast, modern revenue management systems use booking trends, seasonality, room type behavior, and market signals to help automate pricing decisions.

This project is motivated by a real use case involving my dad's company. If successful, this model could help reduce manual pricing work, improve consistency, and potentially save money by providing a lower-cost internal alternative to a commercial revenue management platform. That gives the project both practical business value and real-world impact.

From an academic standpoint, this project also fits the course well because it applies deep learning to structured sequential data. Instead of treating this as a simple regression problem, I want to study how different neural network architectures perform on hotel demand forecasting and whether they can support useful pricing recommendations.

Dataset

The dataset for this project will come from historical hotel data provided by my dad's company. The expected data include reservations, stay dates, booking dates, room types, rates, occupancy, and other operational variables related to pricing decisions. If available, I may also include competitor pricing, booking channel, group bookings, event indicators, and other market-related signals.

This dataset is a good fit for the project because it directly captures the relationship between hotel demand, pricing, and booking behavior over time. If multiple years of data are available, I will use them to capture seasonality, day-of-week effects, holiday effects, and broader demand trends. If room-type-specific or segment-specific data are available, those will also be included.

Methods

Input Data

The input data will be structured as **tabular time-series data**. Each training example will represent either a date or a date-room-type combination, with features describing both the current business situation and recent historical patterns. Likely features include:

- stay date
- booking lead time
- day of week
- month and season
- room type
- current offered rate
- historical occupancy
- historical ADR
- booking pace
- length of stay
- group booking indicators
- event or holiday indicators
- competitor pricing if available
- booking channel or market segment if available

Preprocessing will include handling missing values, encoding categorical variables, normalizing numerical features when needed, and creating lagged or rolling-window features to capture short-term and medium-term trends. Since this is a time-based forecasting problem, I will use a chronological train/validation/test split rather than a random split. Older data will be used for training, a middle block for validation, and the most recent block for testing. If enough data are available, I may also use rolling-origin evaluation.

To support deep learning models, the data will also be organized into sequential windows so that models can look at multiple past days of booking and pricing behavior before predicting future demand.

Target Variable(s)

The main target variable will be a future demand-related regression target, such as:

- rooms sold for a future stay date
- future occupancy percentage
- or expected ADR depending on the available data

The most likely primary target will be future occupancy or rooms sold, since those are directly observable and make the forecasting task more straightforward. After demand is predicted, I will

use that forecast to generate a room pricing recommendation within a realistic range of business constraints.

I may also include a secondary classification task, such as predicting whether a given date falls into a high-demand period.

Model

To better align this project with CSC 561, I plan to compare both classical machine learning baselines and deep learning models.

Baselines

- simple heuristic baseline such as moving average or last-year-same-day demand
- linear regression
- XGBoost or Random Forest for tabular forecasting

Deep Learning Models

- **Multilayer Perceptron (MLP):** a basic feedforward neural network for structured feature prediction
- **LSTM:** to model booking pace and historical occupancy as a sequence over time
- **GRU:** a lighter recurrent alternative to LSTM for time-series forecasting
- **Transformer-based time-series model:** a small Transformer encoder to capture longer-range temporal relationships in booking behavior and demand patterns

The main technical focus of the project is comparing deep learning architectures for hotel demand forecasting using real operational time-series data, while including classical machine learning approaches as baselines.

The overall MVP will likely use a two-stage pipeline:

1. Forecast future demand using one of the candidate models
2. Convert that demand forecast into a room pricing recommendation using a bounded pricing rule or simple optimization layer

This project is not meant to recreate a full commercial revenue management system. The goal is to build a realistic academic MVP that focuses on the core problem of demand forecasting and pricing support.

Metrics and Results

The project will be evaluated using both predictive metrics and business-oriented metrics.

For forecasting performance, I plan to report:

- MAE
- RMSE
- MAPE
- R^2

If I include a classification subtask, I may also report:

- accuracy
- precision
- recall
- F1 score
- AUROC

For business impact, I will compare:

- historical or manual pricing performance
- simulated revenue under model-assisted pricing
- occupancy differences
- ADR differences
- RevPAR differences

Results will be shown using:

- prediction vs. actual plots for occupancy or demand
- time-series plots of recommended price versus actual price
- feature importance or attribution plots
- comparison tables between baselines and deep learning models
- an architecture diagram of the final pipeline
- scenario examples for high-demand weekends, holidays, or local events

If enough data are available, I will also test the model on a held-out later time period to evaluate how well it generalizes to new booking conditions.

Overlap Statement

This project is not intended to overlap with another course project. It is based on a real-world hotel pricing and demand forecasting problem connected to my dad's company, but the model design, implementation, evaluation, and final report will all be completed specifically for CSC 561.